

Verificar se OPIE e MOCANU aparecem

A) "SMART on FHIR" OWASP HIPAA vulnerability

-----Google Scholar (13 resultados, todos abaixo):

Exploring security vulnerabilities in FHIR server implementations: A case study on IBM's FHIR server in the context of the 21st century cures act

CA Opie - 2024

OK

A HIPAA-Aware Agentic AI Co-Pilot Framework: Orchestrating Secure Multi-Step EHR Workflows for Clinical Burden Reduction in US Hospital Systems

MK Damarched - Journal of Drug Delivery & Therapeutics, 2026

Não relevante, foco em diminuir carga de trabalho de médicos

[PDF] CAPF: A Clinical Agent Permission Framework for HIPAA-Aligned Least-Privilege Authorization in Multi-Agent Healthcare AI Systems

R Khanna, J Nandal - Artificial Intelligence in Medicine and ..., 2026

Não relevante, foco em agentes de IA

Zero-trust data fabrics: A policy-driven model for secure cross-cloud healthcare and financial data exchanges

LS Surisetty - International Journal of Advanced Research in ..., 2021

Tangencialmente relevante, identifica saúde e finanças como áreas com dados de alto risco e propõe maneira de melhorar segurança no tráfego de dados entre diferentes nuvens

[PDF] Connected but Compliant-Governance Framework for Interoperable Neurocognitive IoT Systems

MDA RAHAMAN

Tangencial, focado em governança/compliance no contexto de IoT, propõe um framework que gera um score para avaliar o quão seguro é um sistema

AgentCare-Guard: A Capability-Constrained Reference Monitor for Agentic FHIR

Transactions - Adilakshmi Kataru

Parece ser sobre mitigação de ataques via IA, porém "This is a preprint; it has not been peer reviewed by a journal."

[PDF] Security on FHIR

A Marcus - Asymmetrik, Inc., November, 2018

Não é artigo, slides

[HTML] Multi-component pipeline LLMs for interoperable healthcare data: A scoping review from clinical summarization to multimodal integration

JC Ferreira, R Correia, IMD Rosa - 2026 - intechopen.com

Irrelevante, propõe MCP de múltiplas LLMs para unificar dados dispersos em saúde

A HIPAA-compliant web application design framework for next-generation telehealth systems

SS Akib Rahman - International Journal of Research & Technology, 2024 - [ijrt.org](http://ijrt.org)  
Framework para compliance com HIPAA em telemedicina, talvez a parte de segurança seja tangencialmente relevante

zk-ID: A privacy-preserving identity verification framework for healthcare using blockchain, ECDSA, and zero-knowledge proofs

SK Abirami, GR Karpagam... - ... Relevant Applications of ... - [api.taylorfrancis.com](http://api.taylorfrancis.com)

Sobre autenticação em saúde

AI-Driven Cloud-Native Microservices Framework for Secure Healthcare Prescription Automation, Software Reliability, and Scalable Deployment Optimization

SR GUDI - International Journal of Emerging Trends in ..., 2026 -

[internationaljournalssrp.org](http://internationaljournalssrp.org)

Framework com IA para automação de prescrições, irrelevante

Design and Implementation of a Secure Cloud and Network Framework for AI-Based Predictive Analytics in Healthcare and Finance

CJP Hale - International Journal of Computer Technology and ..., 2025 - [ijctece.com](http://ijctece.com)

Arquitetura de rede/cloud com IA, proposta bem parecida com Zero-trust data fabrics

Utilização da Tecnologia SDR para Autenticação de Dispositivos IoMT Baseada em RF Fingerprint

FMM Gomes - 2023 - [search.proquest.com](http://search.proquest.com)

Dissertação de mestrado sobre autenticação de IoT em saúde, focado em como utilizar frequência de rádio para isso

-----IEEE(<https://ieeexplore.ieee.org/Xplore/home.jsp>): 0 resultados

-----ACM (<https://dl.acm.org/>): 0 resultados

-----Springer (<https://link.springer.com/>): 0 resultados

Devido à ausência de resultados em algumas DBs, para verificar como poderia modificar a string de busca fiz uma busca direta por Opie, "Exploring security vulnerabilities in FHIR server implementations", para verificar quais palavras-chave estariam associadas a esse trabalho. Nesse caso, a IEEE retornou 0 resultados e a ACM 48, nenhum desses sendo o artigo de Opie ou mesmo trabalhos sobre FHIR. Talvez tenha a ver com a natureza open source da área (Opie está disponível publicamente no site da sua universidade, por exemplo:

<https://scholarspace.manoa.hawaii.edu/server/api/core/bitstreams/f76a1456-0d4b-472f-85d9-e9ba7fa4069c/content>)

Obs.: portal de periódicos integrados CAPES via proxy unisinos teve 0 resultados  
(<https://www-periodicos-capes-gov-br.ez101.periodicos.capes.gov.br/index.php/acervo/buscaador.html?source=all&q=%22SMART+on+FHIR%22+OWASP+HIPAA+vulnerability>)

Como Opie cita "J. A. Diaz-Rojas, J. O. Ocharan-Hernandez, J. C. Perez-Arriaga, and X. Limon. Web api security vulnerabilities and mitigation mechanisms: A systematic mapping study", usei esse artigo de inspiração para outra string de busca, abaixo:

B) "FHIR" OWASP "API Security" web vulnerabilities HIPAA

-----Google Scholar: 26 resultados, abaixo destaco os que são sobre  
segurança em saúde  
Aparece Opie e alguns outros que já apareceram na string de busca A

ZERO-TRUST SECURITY IN CLOUD API INTEGRATIONS FOR HEALTHCARE SYSTEMS  
R VASA - TPM—Testing, Psychometrics, Methodology in Applied ..., 2025 - tpmmap.org

Managing third-party risk: Framework details for risk management in medical technology  
M Parker, B Partridge, E Svetcov... - Advanced Health ..., 2023 - books.google.com  
Obs.: trecho de livro

Enterprise-Grade Secure API Management using Deep Learning in Healthcare Cloud  
Environments  
R Balamurugan - International Journal of Engineering & Extended ..., 2022 - ijeetr.com

Proactive threat mitigation in API ecosystems through AI-powered anomaly detection  
LS Surisetty - International Journal of Advanced Research in ..., 2023 - ijarcst.org  
Relevante, utilização de machine learning para detectar anomalias de tráfego em APIs, foco  
em segurança de APIs

Cyber-Secure Cloud Architecture Integrating Network and API Controls for Risk-Aware SAP  
Healthcare Data Platforms  
R Sugumar - International Journal of Humanities and Information ..., 2025 - ijhit.info

Cybersecurity Aware Cloud Native AI Framework for Scalable Healthcare Data Analytics via  
APIs  
S Ramakrishna - International Journal of Research Publications in ..., 2024 - ijrpetm.com

Design and Implementation of a Secure Cloud and Network Framework for AI-Based  
Predictive Analytics in Healthcare and Finance  
CJP Hale - International Journal of Computer Technology and ..., 2025 - ijctece.com

Sftsdh: Applying spring security framework with TSD-based oauth2 to protect microservice  
architecture apis  
A Chatterjee, MW Gerdes, P Khatiwada, A Prinz - Ieee Access, 2022 - ieeeexplore.ieee.org  
menciona FHIR  
Trecho sobre experimentos é relevante.

A Secure-by-Design Cloud-Native Framework for Intelligent Healthcare Analytics Integrating Deep Learning and Generative AI for Waste Optimization and ...  
OW Hellström - International Journal of Engineering & Extended ..., 2024 - ijeetr.com

AI-Driven Cloud-Native Microservices Framework for Secure Healthcare Prescription Automation, Software Reliability, and Scalable Deployment Optimization  
SR GUDI - International Journal of Emerging Trends in ..., 2026 - internationaljournalsrp.org

[HTML] FHIR 와 블록체인을 활용한 의료정보 전송의 보안성및 상호운용성 향상 방안  
박용민 - Journal of Digital Contents Society, 2025 - journal.dcs.or.kr parece discutir  
FHIR e HIPAA mas está inteiramente em coreano, teria que traduzir

[PDF] API Security Management  
A Balaganski - KuppingerCole Report, 2025 - files.ziftone.com menciona HIPAA  
Não é paper, mas relatório sobre segurança em APIs no geral por uma firma de analistas  
Em  
/home/rodrigo/Documents/TCC/bibliografia/balaganski-api-security-market-analysis-2025.pdf

[PDF] Enhancing Security and Interoperability of Medical Information Exchange Using FHIR and Blockchain  
YM Park - 디지털콘텐츠학회논문지 (J. DCS), 2025 - journal.dcs.or.kr  
"This study proposes a transmission model that integrates the HL7 FHIR (Fast Healthcare Interoperability Resources) standard with blockchain technology to address security vulnerabilities and limited interoperability in the exchange of patient medical information across healthcare institutions. "  
FOco em blockchain, mas ser em coreano já exclui

[HTML] Multi-component pipeline LLMs for interoperable healthcare data: A scoping review from clinical summarization to multimodal integration  
JC Ferreira, R Correia, IMD Rosa - 2026 - intechopen.com

Secure Software Testing and Validation Frameworks for SAP-Centric Cloud-Native Healthcare Machine Learning Systems Managing PII and Regulated Data  
TA Huber - International Journal of Research Publications in ..., 2026 - ijrpetm.com

Enhancing Medical Device Security: Exploring the Exploitability and Impact of GUI Vulnerabilities Through a Hacking Tool Experiment  
JJK Küfner, S Tahir, ST Bakhsh, L Mohaisen... - ... Conference on Cyber ..., 2023 - Springer  
Testa vulnerabilidades da GUI de dispositivos médicos, irrelevante.

Architectural Frameworks for Secure Inter-Enterprise Integration in Next-Generation Systems  
S Sriramoju - ... Journal of Advanced Engineering Science and ..., 2026 - iadier-academy.org menciona API security, FHIR e HIPAA

Auditoría de ciberseguridad para el análisis de identificación de vulnerabilidades en dispositivo medidor de glucosa  
GA Montenegro Martínez, PV Mosquera Morales... - 2025 - repositorio.uide.edu.ec  
parece ser sobre dispositivo IoT mas inclui por mencionar HIPAA e cybersec

Exclui alguns resultados aparentemente relevantes que eram livros.

-----IEEE(<https://ieeexplore.ieee.org/Xplore/home.jsp>): 0 resultados

-----ACM (<https://dl.acm.org/>): 3 resultados

2 conferências e 1 simpósio, nenhum especificamente sobre saúde, embora as duas conferências ao menos são em segurança, abaixo:

ASIA CCS '26: Proceedings of the ACM Asia Conference on Computer and Communications Security

CCS '25: Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security

-----Springer (<https://link.springer.com/>): 0 resultados

C) "FHIR server" "access control" OR "authentication"      Obs.: muito abrangente, mas scholar retornou alguns resultados aparentemente relevantes

-----Google Scholar: 541 resultados, provavelmente abrangente demais, o que parece ser relevante das primeiras 2 páginas está abaixo

HAPI-FHIR server implementation to enhancing interoperability among primary care health information systems in Sri Lanka: Review of the technical use case

P Jayathissa, R Hewapathirana - arXiv preprint arXiv:2402.02838, 2024 - arxiv.org

Tangencial, "This review underscores the vital role of interoperability in digital health, advocating for a standardized framework. It focuses on implementing a Fast Healthcare Interoperability Resources (FHIR) server, addressing technical, semantic, and process challenges. "

Interoperability and Security in the Implementation of Health Web Applications with SMART on FHIR

LS Santos, GM Martins, FP Itturiet... - J. health ..., 2023 - pesquisa.bvsalud.org

Foco em interoperabilidade e autenticação em IoT

Aparece Opie

**Secure FHIR Transactions Based on the SMART on FHIR Approach in a Health Data Exchange**

**R Ranatunge**, PK Karunapema - Medinfo 2025-Healthcare ..., 2025 - journals.sagepub.com  
Ferramenta de autenticação baseado em FHIR, só descreve a implementação no contexto do Sri Lanka e não discute FHIR no geral, outros sistemas ou APIs.

A smart on fhir conformant infrastructure for recording patient reported outcomes based on open-source components

Y Körber, F Katsch, T Stamm... - ... 2025: Proceedings of ..., 2025 - journals.sagepub.com

Scalable micro-service based approach to FHIR server with golang and No-SQL

FS Shoumik, MIMM Talukder, Al Jami... - ... of Computer and ..., 2017 - ieeexplore.ieee.org  
bem tangencial

Enhancing Healthcare Interoperability with FHIR: A Systematic Approach to Online Data Management

T Kim hoang le, W Kongsiriwattana, A Meny... - Proceedings of the ..., 2024 - dl.acm.org  
parece ser sobre interoperabilidade, não segurança

[PDF] DESIGN AND EVALUATION OF A SECURE CROSS-PLATFORM MEDICAL IMAGE SHARING SYSTEM USING DICOMWEB AND HL7 FHIR STANDARDS

AM Hayes - 2026 - researchgate.net talvez fuja do escopo, não está claro

[PDF] A Scalable and Secured HL7 FHIR Healthcare Platform: Architecture, Load Testing, and Preliminary Findings

V Šafran, U Arioiz, R Sallauka... - International Journal of ..., 2025 - npublications.com

[PDF] Alternative Approaches for Supporting Lattice-based Access Control (LBAC) in the Fast Healthcare Interoperability Resources (FHIR) Standard.

SA Demurjian, T Agresta, E Sanzi, J DeStefano - WEBIST, 2020 - scitepress.org

-----IEEE(<https://ieeexplore.ieee.org/Xplore/home.jsp>): 75,377 resultados

-----ACM (<https://dl.acm.org/>): 35,817 resultados

-----Springer (<https://link.springer.com/>): 149,893 resultados, óbvio que abrangente demais

D) ("FHIR" OR "HL7") AND ("API security" OR "access control") AND ("vulnerability" OR "threat")

-----Google Scholar: 3370 resultados

Aparece Opie mas número de resultados indica que possivelmente é abrangente demais

-----IEEE(<https://ieeexplore.ieee.org/Xplore/home.jsp>): 12 resultados

alguns destacados abaixo

Exploring the Evolving Landscape of API Security Challenges in the Healthcare Industry: A Comprehensive Review

Akram Al-Rumaim;Jyoti D. Pawar

2023 16th International Conference on Security of Information and Networks (SIN)

Justamente o que o nome diz, revisa literatura de segurança de APIs na saúde.

### Measuring Healthcare Data Leaks and Security Flaws at Internet Scale

Nico Brüggemann;Lukas Schmidt;Marvin Dölzer;Marius Brockhoff;Fabian Ising;Christoph Saatjohann;Sebastian Schinzel

2026 IEEE 11th European Symposium on Security and Privacy (EuroS&P)

RELEVANTE E IMPORTANTE, criaram honeypot e geraram dados novos mostrando vulnerabilidades associadas à implementação de HL7 FHIR. Código do honeypot disponível em <https://github.com/FHMS-ITS/Healthcare-Scanning> , detalhes como portas utilizadas disponíveis no apêndice. Apêndice F lista vulnerabilidades CVE encontradas (<https://vulert.com/vuln-db/CVE-2024-51132>)

XML malicioso pode acessar dados sensíveis ou executar código

Metodologia muito bem descrita

### Hybrid Post-Quantum Cryptography (PQC) for Patient-Centric EHR

Edy Prayitno;Noor Alis Setiyadi;Erma Sofiani;Edi Iskandar;Basuki Heri Winarno;Sri Setyowati

2025 8th International Seminar on Research of Information Technology and Intelligent Systems (ISRITI)

Provavelmente irrelevante, não é segurança de APIs, só estratégias de criptografia novas

### A Comprehensive Model for Securing Sensitive Patient Data in a Clinical Scenario

Antonio López Martínez;Manuel Gil Pérez;Antonio Ruiz-Martínez

IEEE Access

Year: 2023

Parece secundário, talvez terciário: menciona cybersec em saúde mas nada de APIs, OWASP, testes ou dados novos

### Practical and Efficient Non-Interactive Multi-Client Searchable Encryption for E-Healthcare Systems

Yanpeng Ba;Yuan Ping;Xiali Hei;Zheng Yuan

IEEE Transactions on Emerging Topics in Computing

Year: 2026

Focado em criptografia, "we propose a non-interactive multi-client dynamic searchable symmetric encryption (NIMC-DSSE) scheme tailored to the e-healthcare context. "

### Securing Mobile e-Health Environments by Design: A Holistic Architectural Approach

Helder Aranha;Massimiliano Masi;Tanja Pavleska;Giovanni Paolo Sellitto

2019 International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob)

Irrelevante

-----ACM (<https://dl.acm.org/>): 87 resultados    abaixo alguns destaques

October 2023 Published By ACM

Ensuring FHIR Authentication and Data Integrity by Smart Contract and Blockchain Enabled NFT - Ching-Nung Yang, Hsin-Chuan Kuo, Hsiang-Han Cheng

Irrelevante. " we use (F)HIR and (E)thereum smart contract to ensure data Integrity in (E)MR (FEE)"

May 2026

Published By ACM

### **Benchmarking Security Features in Open-Source Healthcare Software Systems**

**YMS Tharaka,** WVC Maduwanthi, L. Munasinghe

Relevante, mas foco em Electronic Health Records (prontuários). Talvez útil para justificar necessidade de FHIR. Mas sobretudo mostra quais sistemas falha em qual aspecto (nenhum criptografa payloads FHIR, todos dependem de TLS). Fala sobre como setor de saúde viu grande aumento de phishing, estima-se que informação médica vale de 10-40 vezes mais que de cartão de crédito no mercado negro. "Healthcare-specific security requirements include administrative safeguards (security policies, training), physical safeguards (facility access controls, device controls), and technical safeguards (access control, audit logs, encryption) as mandated by HIPAA Security Rule [26]." Matriz comparativa similar à ideia de FHIR x HIPAA x OWASP, mesmo que tópico distinto "However, REST and SOAP API endpoints, along with HL7 and FHIR messages, rely entirely on transport layer security without application-level encryption or message-level protection." Tabelas abaixo (2 de 6)

**Table 1: Comparative Security Benchmarking of Open-Source EMR Systems**

| EMR System | Authentication          | Encryption                      | Interoperability Security    | RBAC Security                 | Compliance Adoption               |
|------------|-------------------------|---------------------------------|------------------------------|-------------------------------|-----------------------------------|
| OpenEMR    | Strong (2FA, policies)  | Moderate (TLS, DB configurable) | Moderate (HL7, limited FHIR) | Strong (fine-grained modules) | Strong (HIPAA docs, admin config) |
| OpenMRS    | Moderate (modular LDAP) | Moderate (TLS, modular)         | Strong (HL7, FHIR, OpenHIM)  | Strong (granular, modular)    | Moderate (HIPAA/GDPR via modules) |
| GNU Health | Weak (basic login)      | Weak (TLS only)                 | Moderate (HL7, DICOM)        | Moderate (module-level)       | Weak (manual compliance)          |
| OSCAR EMR  | Strong (2FA, policies)  | Moderate (TLS, DB configurable) | Moderate (HL7, FHIR plugins) | Strong (fine-grained modules) | Strong (HIPAA support)            |
| Bahmni     | Moderate (modular LDAP) | Moderate (TLS, modular)         | Strong (HL7, FHIR, OpenHIM)  | Strong (granular, modular)    | Moderate (inherits OpenMRS)       |

**Table 2: Authentication Features in Open-Source EMR Systems**

| Authentication Feature                     | OpenEMR                     | OpenMRS                   | GNU Health             | OSCAR EMR                           | Bahmni                         |
|--------------------------------------------|-----------------------------|---------------------------|------------------------|-------------------------------------|--------------------------------|
| Username and Password Authentication       | Supported                   | Supported                 | Supported              | Supported                           | Supported                      |
| Password Hashing Algorithm                 | bcrypt                      | Salted and configurable   | Basic hashing          | SHA-256                             | Salted hashing (OpenMRS-based) |
| Multi-Factor Authentication (MFA / 2FA)    | Native support (TOTP-based) | Available through modules | Not supported natively | Native support (TOTP-based)         | Available through modules      |
| LDAP / Active Directory Integration        | Supported                   | Supported via modules     | Limited support        | External or proxy-based integration | Supported via modules          |
| OAuth2 / OpenID Connect (Single Sign-On)   | Supported                   | Supported via modules     | Not supported          | Supported via external integration  | Supported                      |
| Session Timeout and Auto Logout            | Configurable and enforced   | Configurable              | Basic session handling | Configurable and enforced           | Configurable                   |
| Password Policy Enforcement                | Strong                      | Moderate                  | Basic                  | Strong (complexity and expiration)  | Moderate                       |
| Account Lockout and Brute-force Protection | Supported                   | Available through modules | Not supported          | Supported                           | Available through modules      |

Tabela 4, abaixo, fala de interoperabilidade, logo FHIR

Table 4: Interoperability Security Features in Open-Source EMR Systems

| Interoperability Feature                                  | Security | OpenEMR                 | OpenMRS              | GNU Health           | OSCAR EMR                     | Bahmni          |
|-----------------------------------------------------------|----------|-------------------------|----------------------|----------------------|-------------------------------|-----------------|
| HL7 Messaging Support                                     |          | Supported               | Supported            | Supported            | Supported                     | Supported       |
| FHIR API Support                                          |          | Limited support         | Supported            | Limited support      | Supported via plug-ins        | Supported       |
| DICOM Integration                                         |          | Limited                 | Limited              | Supported            | Limited                       | Limited         |
| API Authentication Mechanism                              |          | OAuth2 / OpenID Connect | OAuth2 (via modules) | Basic authentication | OAuth2 (via plugins or proxy) | OAuth2          |
| Token-Based Access Control                                |          | Supported               | Supported            | Limited              | Supported                     | Supported       |
| Transport Layer Security for Interoperability             |          | Secured via TLS         | Secured via TLS      | Secured via TLS      | Secured via TLS               | Secured via TLS |
| Message-Level Encryption (HL7 / FHIR payloads)            |          | Not supported           | Not supported        | Not supported        | Not supported                 | Not supported   |
| Message Integrity and Digital Signatures                  |          | Not supported           | Not supported        | Not supported        | Not supported                 | Not supported   |
| External Interoperability Gateway Support (e.g., OpenHIM) |          | Limited                 | Supported            | Not supported        | Not supported                 | Supported       |
| Fine-Grained API Scope Control                            |          | Moderate                | Strong               | Weak                 | Moderate                      | Strong          |

“HL7 and FHIR payloads transmitted between systems depend entirely on TLS confidentiality protection, creating exposure risks in multi-hop network scenarios or when intermediaries terminate encryption.”

“but all systems require substantial administrative effort for full compliance.” Talvez seja com isso que eu possa contribuir

April 2024

Published By ACM

**Privacy Preservation of Electronic Health Records in the Modern Era: A Systematic Survey**

**Raza Nowrozy**, Khandakar Ahmed, A. S. M. Kayes, Hua Wang, Timothy R. McIntosh

Bem genérico, no máximo secundário. Com isso dito, a metodologia é bem feita e isso pode ser referência.

"none of the existing studies tested their proposed method using either real samples or raw data of EHRs, which puts the external validity of these studies in question."

Prova a necessidade de trabalhos mais empíricos.

January 2016

Published By ACM

A cloud security framework for a data centric WSN application

Sayantani Saha, Rounak Das, Suman Datta, Sarmistha Neogy

Bem genérico, mas ser de 2016 já exclui.

August 2026

Published By ACM

Computing-Empowered E-Health and Telemedicine: System Architecture, Core Algorithms, and System Challenges--Submission Template for ACM Papers

Author Yucong Wan

Irrelevante e vários problemas. "Submission Template for ACM Papers", " Beijing Huizhong Bokang Sports Technology" não parece ser uma universidade real

October 2010 Published By ACM XML security in healthcare web systems      Author

PictureWasim A. Al-Hamdani      antigo demais, excluído

July 2026

JUST ACCEPTED

Published By ACM

Towards Secure Healthcare 4.0: Zero-Trust Based Continuous Authentication Framework Using Biometric Pulse Data

Author Bhanu Chander

Irrelevante, criptografia para IoT

November 2025

Published By ACM

A Distributed Architecture for Privacy-Preserving Management, Authorization, and Secure Update of Genomic Data

Author Yunyan Xiong, Author PictureYijun Mao, Author PictureMingju Yu, Author

PictureXiaolan Zhang

Irrelevante, só cripto. "This paper outlines a distributed architecture for the de-identification, authorization, and secure updating of genomic data."

June 2026

Published By ACM

A Systematic Review of Intrusion Detection Systems for Internet of Medical Things: Performance, Efficiency, Explainability, and Generalization

Author Oswald Adohinzin, Author PictureYoussef Harrath

Faz o que o nome diz, irrelevante para mim.

-----Springer (<https://link.springer.com/>): 304 resultados, talvez abrangente demais

Vários resultados com blockchain e IoT em saúde, mas no geral tangentes

Exemplo:

Naresh, V.S., Sada, R., Allu, R.J.P. et al. Exploring the potential of blockchain technology in modern healthcare systems. *Peer-to-Peer Netw. Appl.* 18, 314 (2025).  
<https://doi.org/10.1007/s12083-025-02132-3>